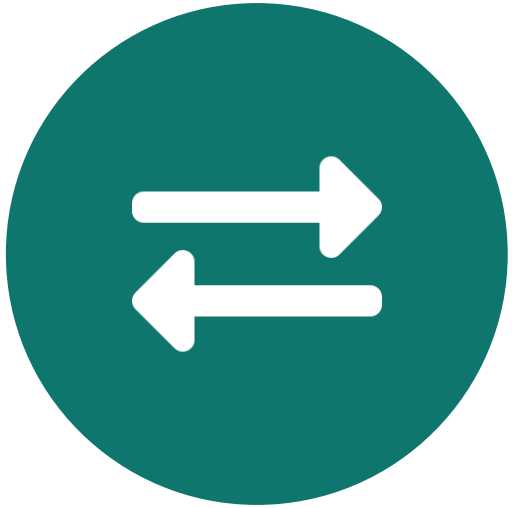
HEXADECIMAL CONVERSIONS

Hexadecimal → Decimal

Hexadecimal → Binary

Hexadecimal → Octal



Hexadecimal Number System

Definition:

Hexadecimal uses 16 symbols. One hex digit represents exactly 4 binary bits.

Widely used in memory addresses, color codes, and debugging.

Digits & Characters:

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

A=10 B=11 C=12 D=13 E=14 F=15

Base (Radix): **16**

Representation:

$(177)_{16}$ $(FF)_{16}$ $(17D)_{16}$ $(2D.4)_{16}$

Why does Hexadecimal use 4 bits?

Hexadecimal has 16 digits (base 16):

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

3 bits

$2 \times 2 \times 2$

= 8 patterns

Not enough for 0 to F

4 bits

$2 \times 2 \times 2 \times 2$

= 16 patterns

Exactly 16 digits: 0 to F

5 bits

$2 \times 2 \times 2 \times 2 \times 2$

= 32 patterns

More than needed

Base 16 = 2^4 → 1 hex digit = 4 bits

How to make the 4-bit table

$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7

$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10 = A
1	0	1	1	11 = B
1	1	0	0	12 = C
1	1	0	1	13 = D
1	1	1	0	14 = E
1	1	1	1	15 = F

Start from 0 and keep adding 1. The right bit changes every row, the next every 2 rows, then every 4 rows, and the left bit every 8 rows. Digits 10 to 15 are written as A to F.

Hexadecimal → Decimal | $(1A)_{16}$

Without point

Position : 1 0

Hex : 1 A

Value:

$1\times(16)^1 + 10\times(16)^0$

$$16 + 10 = (26)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Hexadecimal → Decimal: Method

1. Write the place value (power of 16) under every hex digit.
2. Change each hex letter to its value (A = 10 … F = 15) and multiply by its place value.
3. Add all the results to get the decimal number.

$$\text{Decimal} = \Sigma\,(\,\text{digit} \times 16^{\text{position}}\,)$$

Hexadecimal → Decimal | $(2F)_{16}$

Without point

Position :	1	0
Hex :	2	F

Value:

$2\times(16)^1 + 15\times(16)^0$

$$32 + 15 = (47)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Hexadecimal → Decimal | $(A.8)_{16}$

With point

Position :	0	.	−1
Hex :	A	.	8

Value:

$10\times(16)^{0}$ + $8\times(16)^{-1}$

$$10 + 0.5 = (10.5)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Hexadecimal → Binary | $(2F)_{16}$

Without point



4-bit table (the values used in this example are highlighted)

0 **0000**	1 **0001**	2 **0010**	3 **0011**	4 **0100**	5 **0101**	6 **0110**	7 **0111**
8 **1000**	9 **1001**	A = 10 **1010**	B = 11 **1011**	C = 12 **1100**	D = 13 **1101**	E = 14 **1110**	F = 15 **1111**

Each hex digit becomes 4 bits. Grey zeros at the start are dropped.

Answer: $(101111)_2$

Hexadecimal → Binary: Method

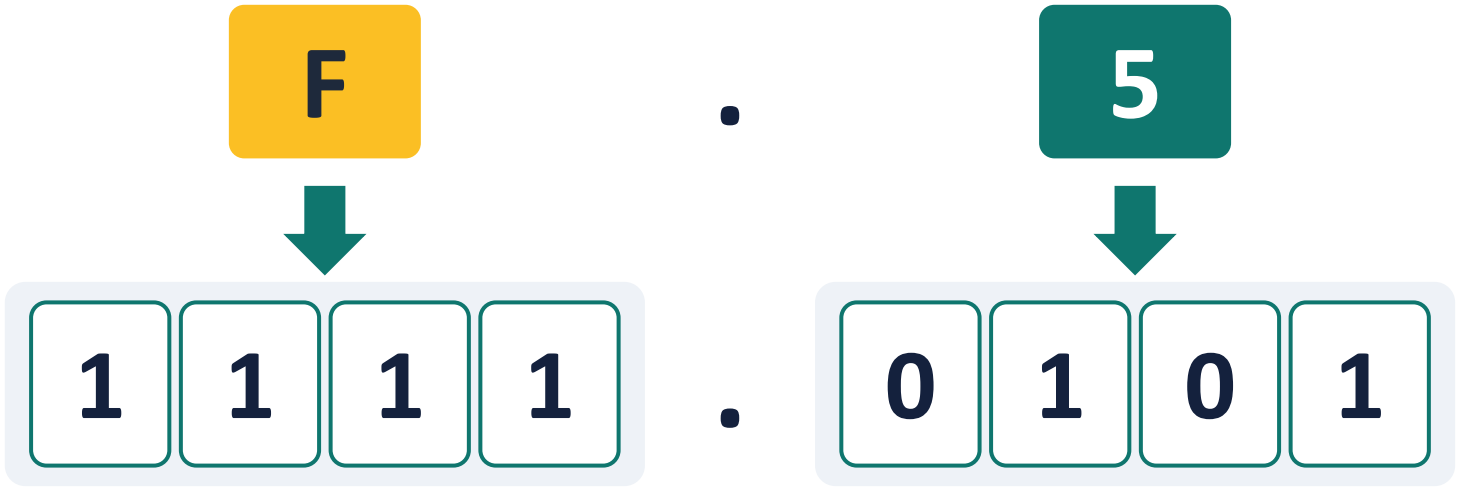
1. Take every hex digit separately.
2. Write the 4-bit binary code of each digit (4-bit table made earlier).
3. Join all the groups and drop the leading zeros.

Lookup table: hex digit → 4 bits (= decimal value)

0 **0000**	1 **0001**	2 **0010**	3 **0011**	4 **0100**	5 **0101**	6 **0110**	7 **0111**
8 **1000**	9 **1001**	A = 10 **1010**	B = 11 **1011**	C = 12 **1100**	D = 13 **1101**	E = 14 **1110**	F = 15 **1111**

Hexadecimal → Binary | $(F.5)_{16}$

With point



4-bit table (the values used in this example are highlighted)

0 **0000**	1 **0001**	2 **0010**	3 **0011**	4 **0100**	5 **0101**	6 **0110**	7 **0111**
8 **1000**	9 **1001**	A = 10 **1010**	B = 11 **1011**	C = 12 **1100**	D = 13 **1101**	E = 14 **1110**	F = 15 **1111**

Each hex digit becomes 4 bits.

Answer: $(1111.0101)_2$

Why does Octal use 3 bits?

Octal has 8 digits (base 8):

0	1	2	3	4	5	6	7

2 bits

2×2

= 4 patterns

Not enough for 0 to 7

3 bits

$2 \times 2 \times 2$

= 8 patterns

Exactly 8 digits: 0 to 7

4 bits

$2 \times 2 \times 2 \times 2$

= 16 patterns

More than needed

Base 8 = 2^3 → 1 octal digit = 3 bits

How to make the 3-bit table

$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Digit
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

2^0 column

0 1 0 1 0 1 0 1

changes in every row

2^1 column

0 0 1 1 0 0 1 1

changes after every 2 rows

2^2 column

0 0 0 0 1 1 1 1

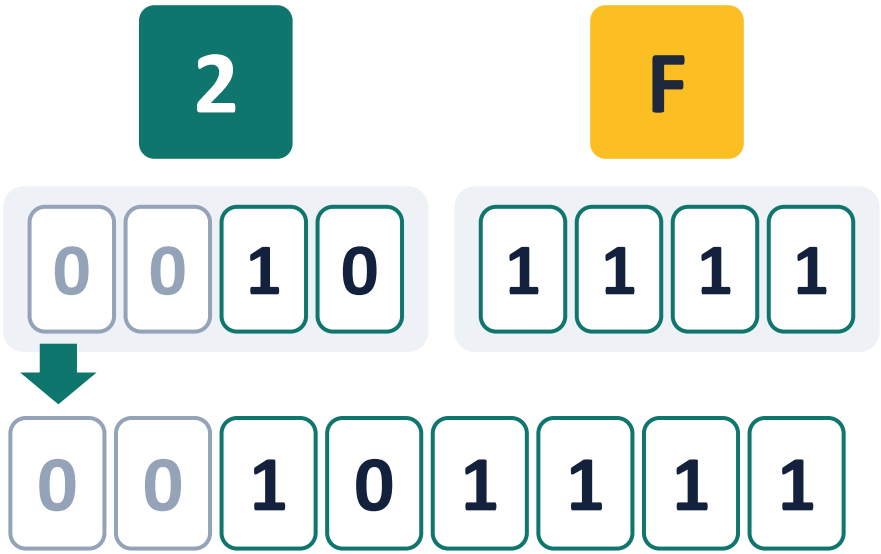
changes after every 4 rows

Hexadecimal → Octal | $(2F)_{16}$

Without point · Step 1 of 2

Hexadecimal

Binary (4 bits each)



Join all the bits

4-bit table (the values used in this example are highlighted)

0 **0000**	1 **0001**	2 **0010**	3 **0011**	4 **0100**	5 **0101**	6 **0110**	7 **0111**
8 **1000**	9 **1001**	A = 10 **1010**	B = 11 **1011**	C = 12 **1100**	D = 13 **1101**	E = 14 **1110**	F = 15 **1111**

Grey zeros are dropped in the next step.

Hexadecimal → Octal: Method

1. Convert every hex digit into 4 bits.
2. Join all the bits, then make groups of 3 bits from the binary point.
3. Replace each group of 3 bits with its octal digit.

Lookup table: 3 bits → octal digit

000	001	010	011	100	101	110	111
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**

Hexadecimal → Octal | $(F.5)_{16}$

With point · Step 1 of 2

Hexadecimal F . 5

Binary (4 bits each) 1 1 1 1 . 0 1 0 1

Join all the bits 1 1 1 1 . 0 1 0 1

4-bit table (the values used in this example are highlighted)

0 0000	1 0001	2 0010	3 0011	4 0100	5 0101	6 0110	7 0111
8 1000	9 1001	A = 10 1010	B = 11 1011	C = 12 1100	D = 13 1101	E = 14 1110	F = 15 1111